

Algorithm 1: Self-Supervised Primal-Dual Training Loop

Input: Network active/reactive profiles $\mathbf{P}_d, \mathbf{Q}_d$, epochs K_{primal} and K_{dual} , penalty ρ

Output: Updated models $\text{ACPFPrimalGAT}_{\theta}$, $\text{ACPFDualGAT}_{\phi}$, violation $\mathcal{V}_{\text{max}}$

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1 for  $iter = 1$  to  $K_{\text{primal}}$  do
2   for  $minibatch (\mathbf{P}_m, \mathbf{Q}_m)$  do
3     Estimate grid state:  $\mathbf{x} = [\mathbf{P}_g, \mathbf{Q}_g, \mathbf{V}, \boldsymbol{\theta}] =$ 
        $\text{ACPFPrimalGAT}_{\theta}(\mathbf{P}_m, \mathbf{Q}_m)$ ;
4     Evaluate multipliers
        $\boldsymbol{\lambda} = \text{ACPFDualGAT}_{\phi}(\mathbf{P}_m, \mathbf{Q}_m)$  without
       gradient tracking;
5     Compute residuals:
6      $\mathbf{r}(\mathbf{x}) = \text{PhysicsLayer}(\mathbf{x}, \mathbf{Y}_{\text{bus}}, \mathbf{P}_m, \mathbf{Q}_m)$ ;
7     Compute the primal augmented Lagrangian:
8      $\mathcal{L}_{\text{primal}} = \mathbb{E} [\boldsymbol{\lambda}^{\top} \mathbf{r}(\mathbf{x}) + \frac{\rho}{2} \|\mathbf{r}(\mathbf{x})\|_2^2]$ ;
9     Update primal weights  $\boldsymbol{\theta}$  by gradient descent;
10  end
11 end
12 Freeze the dual reference checkpoint:  $\phi_{\text{old}} \leftarrow \phi$ ;
13 Generate frozen multipliers:
    $\boldsymbol{\lambda}_{\text{old}} = \text{ACPFDualGAT}_{\phi_{\text{old}}}(\mathbf{P}_d, \mathbf{Q}_d)$ ;
14 for  $iter = 1$  to  $K_{\text{dual}}$  do
15   for  $minibatch (\mathbf{P}_m, \mathbf{Q}_m)$  do
16     Form targets:  $\boldsymbol{\lambda}_{\text{target}} = \boldsymbol{\lambda}_{\text{old},m} + \rho \cdot \mathbf{r}(\mathbf{x}_{\text{detach}})$ ;
17     Evaluate  $\boldsymbol{\lambda}_{\phi} = \text{ACPFDualGAT}_{\phi}(\mathbf{P}_m, \mathbf{Q}_m)$ ;
18     Update  $\phi$  by minimizing
        $\mathcal{L}_{\text{dual}} = \frac{1}{2} \|\boldsymbol{\lambda}_{\phi} - \boldsymbol{\lambda}_{\text{target}}\|_2^2$ ;
19   end
20 end
21 Evaluate the largest residual:
22  $\mathcal{V}_{\text{max}} = \max (\|\mathbf{r}_P(\mathbf{x})\|_{\infty}, \|\mathbf{r}_Q(\mathbf{x})\|_{\infty})$ ;
23 Update the penalty:  $\text{maybe\_update\_rho}(\mathcal{V}_{\text{max}})$ ;
24 return  $(\mathcal{L}_{\text{primal}}, \mathcal{V}_{\text{max}})$ ;
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